```
.data
        msg1: .string "Enter your roll number: "
        msg2: .string "The sum from 1 to your roll number is: "

.text
.globl main

 main:
        # Print input statement
        li, a7, 4
        la a0, msg1
        ecall

        # Save input number
        li a7, 5
        ecall
        mv t0, a0

        li t1, 1
        li t2, 0

LOOP:
        bgt t1, t0, EXIT

        add t2, t2, t1  # sum = sum + i
        addi t1, t1, 1  # i = i + 1

        j LOOP

EXIT:
        li, a7, 4
        la a0, msg2
        ecall

        mv a0, t2
        li a7, 1
        ecall

        li a7, 10
        ecall
```

```
Enter your roll number: 38
The sum from 1 to your roll number is: 741
```

```
.data
        n: .string "Enter sum of your roll number (n): "
        ans: .string "The factorial is: "

.text
.globl main

main:
        # Display input message
        li a7, 4
        la a0, n
        ecall

        # Input integer
        li a7, 5
        ecall
        mv t1, a0       # factorial condition

        # t1 is our counter
        li t2, 1        # Answer storing
        li t3, 1
LOOP:
         blt t1, t3, EXIT

         mul t2, t2, t1
         addi t1, t1, -1

         j LOOP

EXIT:
        li, a7, 4
        la a0, ans
        ecall

        mv a0, t2
        li a7, 1
        ecall

        li a7, 10
        ecall
```

```
Enter sum of your roll number (n): 11
The factorial is: 39916800
```

```
.data
        n: .string "Enter some integer "
        composite: .string "The number is composite"
        prime: .string "The number is prime"
        neither: .string "The number is neither prime nor composite"

.text
.globl main

main:
        # Display input message
        li a7, 4
        la a0, n
        ecall

        # Input integer
        li a7, 5
        ecall
        mv t0, a0       # factorial condition

        li t1, 2        # Initiate for loop and check
        blt t0, t1, IS_NEITHER

LOOP:
        mul t2, t1, t1          # t2 = i * i
        bgt t2, t0, IS_PRIME    # t2 > n = PRIME

        rem t3, t0, t1          # t3 = n % i
        beq t3, zero, IS_COMPOSITE

        addi t1, t1, 1          # i++
        j LOOP

IS_PRIME:
        li, a7, 4
        la a0, prime
        ecall
        j EXIT

IS_COMPOSITE:
        li, a7, 4
        la a0, composite
        ecall
        j EXIT

IS_NEITHER:
        li, a7, 4
        la a0, neither
        ecall

EXIT:
        li a7, 10
        ecall
```

```
Enter some integer 37
The number is prime
```

```
Enter some integer 38
The number is composite
```

```
.data
    msg1: .asciz "Enter your roll number: "
    msg2: .asciz "Enter your friend's roll number: "
    result: .asciz "The approximate average is: "

.text
.globl main

main:
    # Get your roll number
    li a7, 4
    la a0, msg1
    ecall

    li a7, 5
    ecall
    mv s0, a0

    # Get friend's roll number
    li a7, 4
    la a0, msg2
    ecall

    li a7, 5
    ecall
    mv s1, a0

    mv a0, s0           # Parameter 1: Your roll number
    mv a1, s1           # Parameter 2: Friend's roll number
    li a2, 3            # Parameter 3: The number 3

    #AVERAGE procedure call
    jal AVERAGE

    #Store returned result
    mv s2, a0

    # 6. Display result
    li a7, 4
    la a0, result
    ecall

    mv a0, s2
    li a7, 1
    ecall

    # 7. Exit
    li a7, 10
    ecall

AVERAGE:
    # Sum the three parameters
    add t0, a0, a1
    add t0, t0, a2

    div a0, t0, a2

    ret
```

```
Enter your roll number: 38
Enter your friend's roll number: 39
The approximate average is: 26
```

```
.data
    num: .asciz "Enter the base (Sum of your roll number): "
    pow: .asciz "Enter the power (0-9): "
    result: .asciz "The result of num^pow is: "

.text
.globl main

main:
    #Input for num
    li a7, 4
    la a0, num
    ecall
    li a7, 5
    ecall
    mv s0, a0

    #Input for pow
    li a7, 4
    la a0, pow
    ecall
    li a7, 5
    ecall
    mv s1, a0

    #POWER function
    mv a0, s0
    mv a1, s1
    jal POWER

    mv s2, a0            # Save result

    #Result
    li a7, 4
    la a0, result
    ecall

    mv a0, s2
    li a7, 1
    ecall

    li a7, 10
    ecall

POWER:
    li t0, 1             #result storing
    li t1, 0             #counter

POWER_LOOP:
    beq t1, a1, EXIT

    mul t0, t0, a0       # result = result * base
    addi t1, t1, 1       # i++

    j POWER_LOOP

EXIT:
    mv a0, t0
    ret
```

```
Enter the base (Sum of your roll number): 11
Enter the power (0-9): 2
The result of num^pow is: 121
```